

Magnum Opus Clinic

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Problem statement

Danish General Practitioners (GPs) have contact with around 49 patients per day on average. Operating with such a high volume of patients carries risks of making mistakes across three core steps: diagnosis, prescribing, and referral.

Source: Beskrivelse af almen praksissektoren i Danmark (2016)

Project's goal

The goal of this year's semester project is to identify potential mistakes during the consultation made by GPs by providing an AI-assisted workflow, where both the medical AI and the GP can look for mistakes made during the consultation.

Proposed Solution

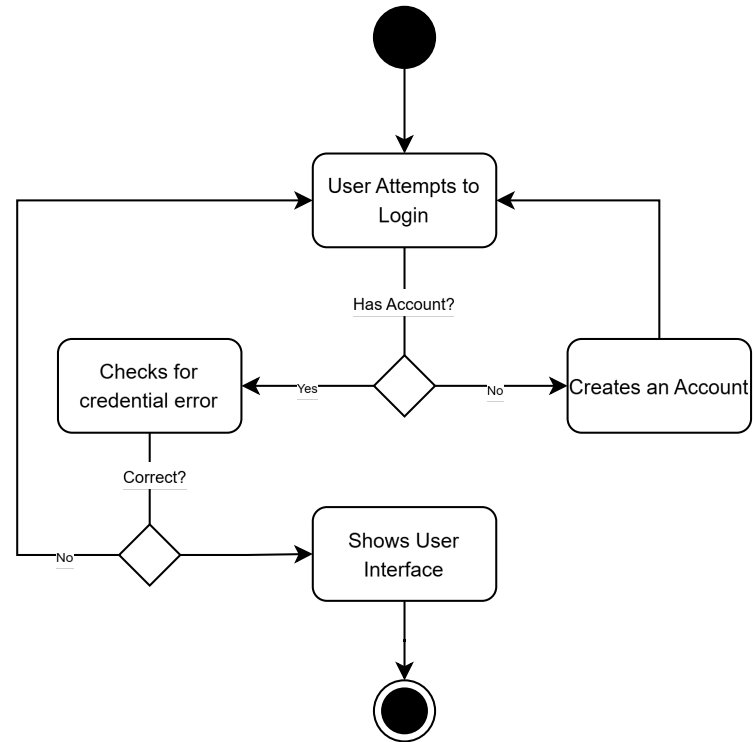
- Locally hosted OPD management system for workflow automation
- Isolated services, component-based architecture
- Privacy-first approach

Core Features

- Appointment booking
- Speech-To-Text
- LLM summarisation
- Prescription generation
- LLM suggestions
- Prescription review
- PDF generation, email sending

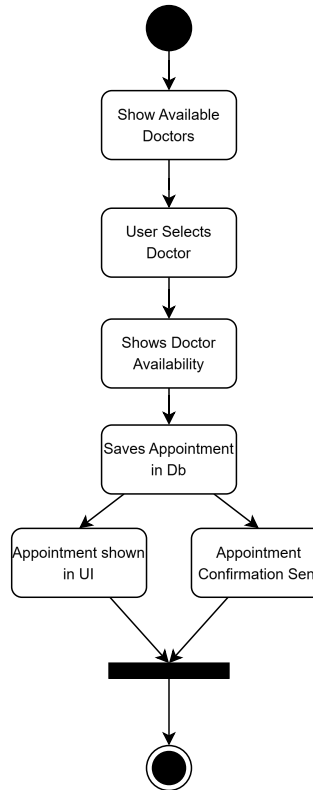
System Workflow - Login

Login - User Workflow

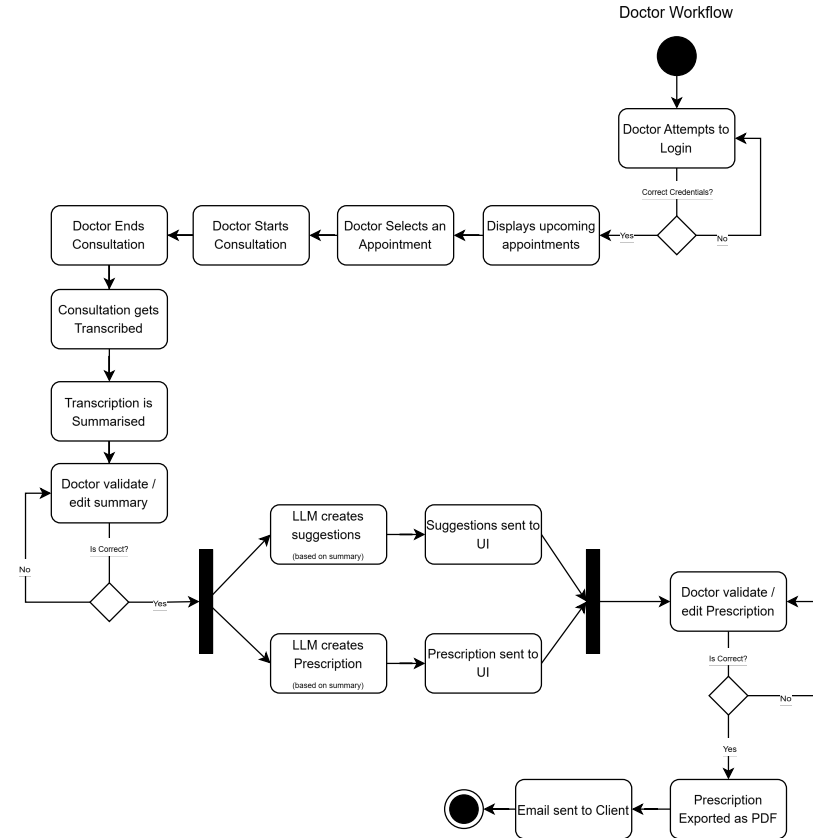


System Workflow - Booking

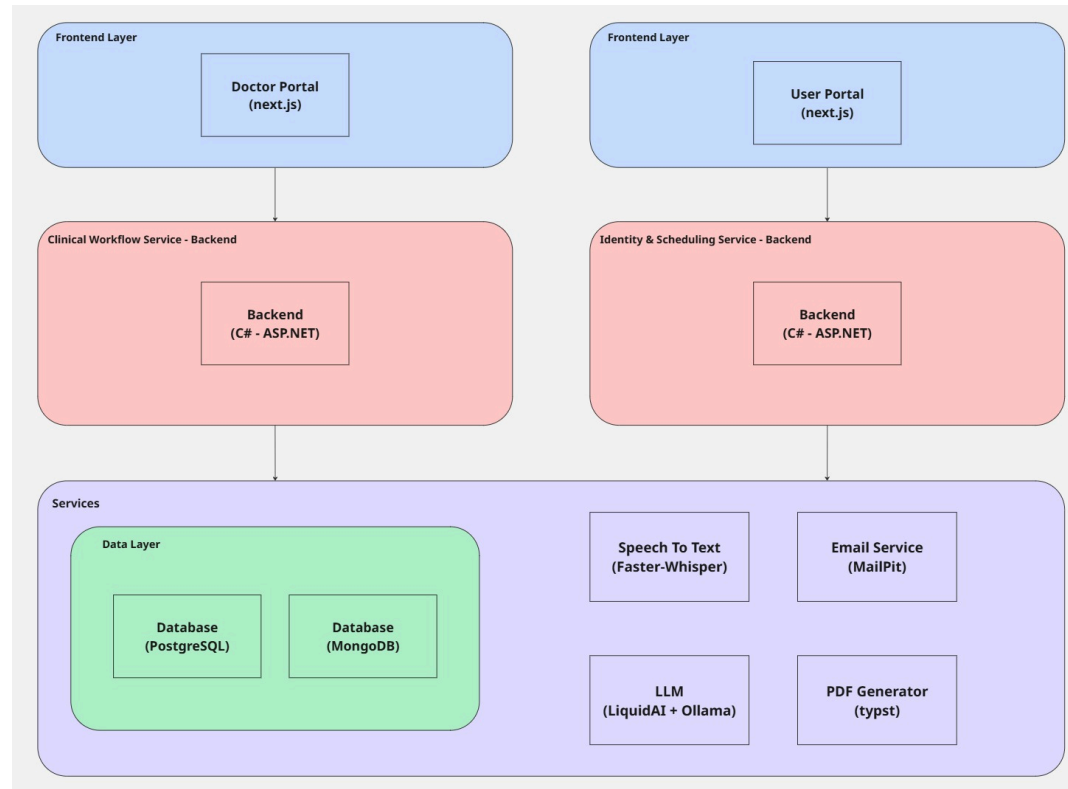
Booking - User Workflow



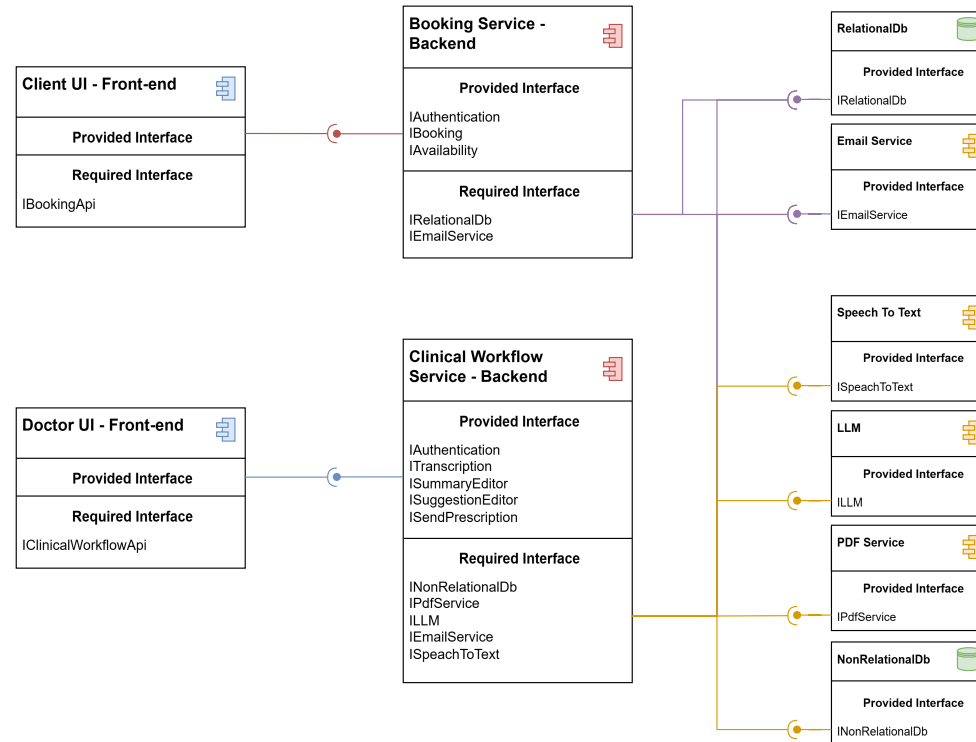
System Workflow - Consultation



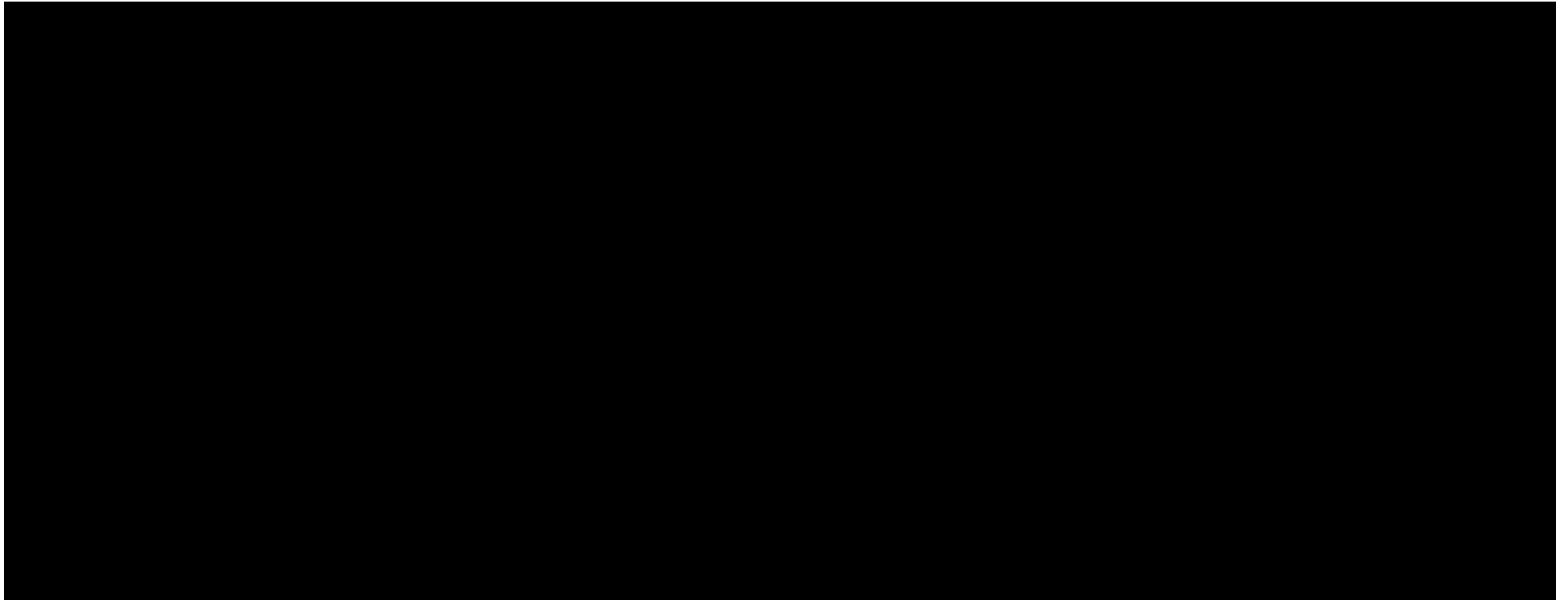
Implied Tech Stack



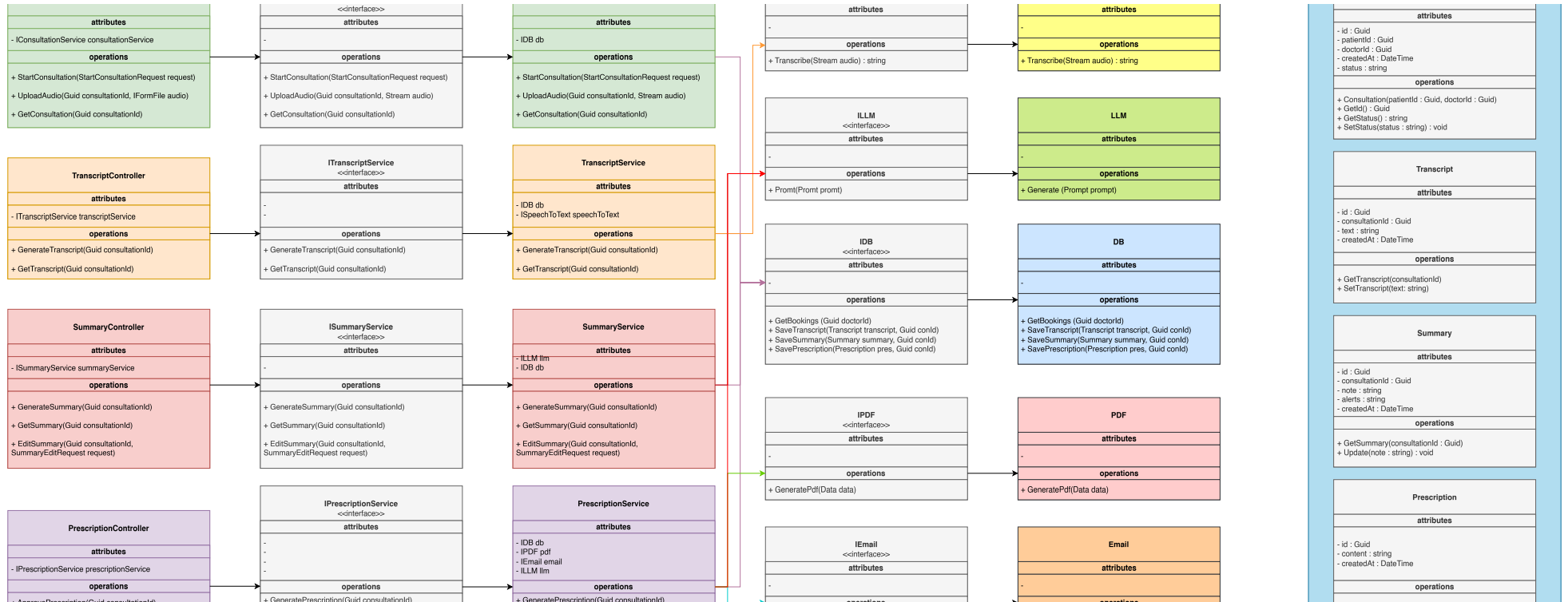
System Architecture



Database Schema



Class Structure



Current Status

- Both backends fully functional (booking + consultation)
- All 4 infrastructure services running (STT, LLM, PDF, Email)
- Both databases up (PostgreSQL + MongoDB)
- Cleaning up & aligning both backends
- Updating diagrams to match the implementation
- Implementing authentication & authorisation

Challenges / Blockers

- Coming up with a working component-based system design
- Model parameter size limitations, due to low performance hardware
- Broad project scope/specification

Left to Do

- Build patient portal frontend (Next.js)
- Build doctor portal frontend (Next.js)
- Unit tests and functional tests (target: $\geq 80\%$ coverage)

Left to Do

Dear XY, here is the breakdown of the consultation:

Symptoms

- Lorem ipsum dolor sit amet, consectetur adipiscing elit.
- Sed id orci ac massa porta fermentum.
- Sed varius nibh at lectus vehicula, ac pharetra purus varius.
- Morbi tempus odio vitae maximus condimentum.
- Suspendisse quis lacus a sapien sodales pulvinar.
- Pellentesque bibendum ante sed est sollicitudin, in iaculis libero dignissim.

AI Suggestions

- Nulla varius nibh et sagittis ultrices.
- Praesent ac lectus et felis fermentum gravida.
- Cras quis nulla ut tortor consectetur aliquet porta nec magna.
- Suspendisse posuere sapien a malesuada finibus.
- Suspendisse dignissim neque at risus porttitor, in ultrices sapien imperdiet.
- Praesent ut nibh vel sem laoreet convallis.

Diagnosis

- Donec fermentum ante sed sapien finibus, eu luctus purus volutpat.

Advice/Prescription

- Nulla varius nibh et sagittis ultrices.
- Praesent ac lectus et felis fermentum gravida.
- Cras quis nulla ut tortor consectetur aliquet porta nec magna.
- Suspendisse posuere sapien a malesuada finibus.

Seal consultation

Extra (If We Have Time)

- Stress testing & edge case testing
- Benchmark testing (comparing LLM model performance)
- Switchable LLM models / try alternative models
- Diagnostic image upload during consultations

Wrapping up

- Our team made great progress
- We have a thought-out roadmap
- We believe our software will be a great help for doctors

The End